New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

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Business Questions

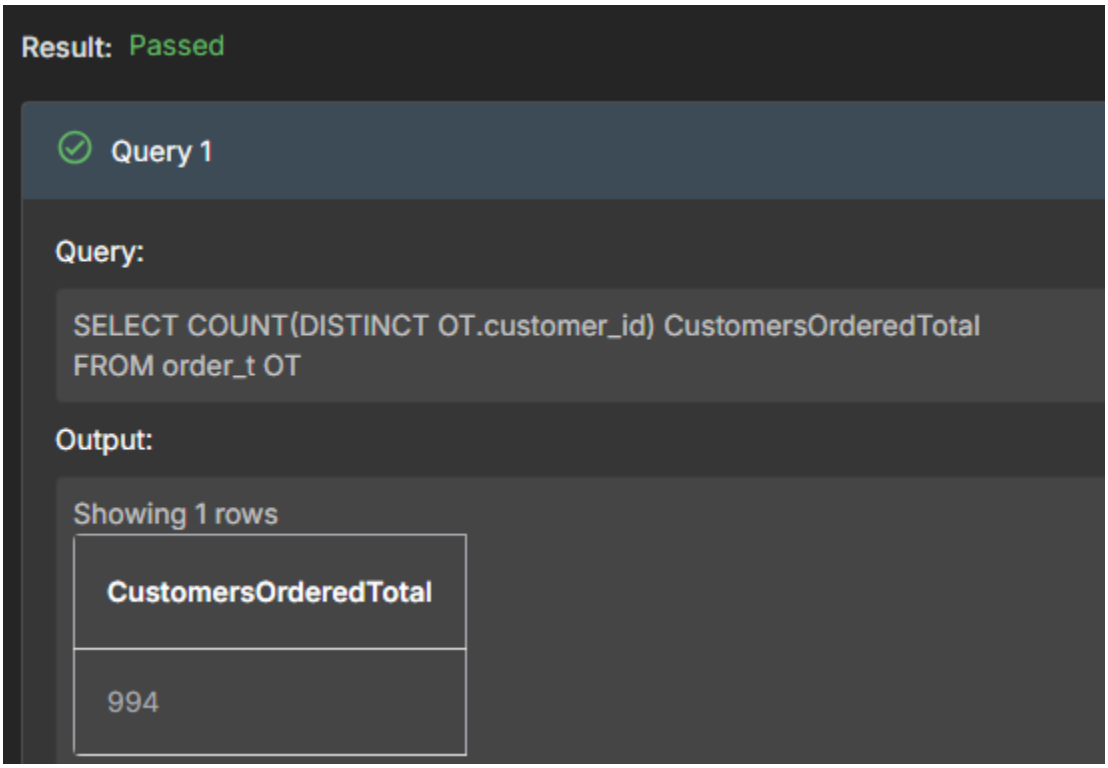
Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
SELECT COUNT(DISTINCT OT.customer_id) CustomersOrderedTotal
FROM order_t OT;
```

```
SELECT CT.state,
       COUNT(DISTINCT OT.customer_id) StateTotal
FROM customer_t CT
    LEFT JOIN order_t OT USING(customer_id)
GROUP BY CT.state
ORDER BY StateTotal DESC;
```

Output:



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Result: Passed

Query 1

Query:

```
SELECT CT.state,
      COUNT(DISTINCT OT.customer_id) StateTotal
FROM customer_t CT
   LEFT JOIN order_t OT USING(customer_id)
GROUP BY CT.state
ORDER BY StateTotal DESC
```

Output:

Showing first 10 rows out of 49 rows

state	StateTotal
Texas	97
California	97
Florida	86
New York	69
District of Columbia	35

Observations and Insights:

- California and Texas have the most customers that placed orders
- The number of customers per state drops significantly after the top 4 states

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Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT PT.vehicle_maker, SUM(OT.quantity) VehicleCount
FROM order_t OT
    JOIN product_t PT USING(product_id)
GROUP BY PT.vehicle_maker
ORDER BY VehicleCount DESC
LIMIT 5;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT PT.vehicle_maker, SUM(OT.quantity) VehicleCount
FROM order_t OT
   JOIN product_t PT USING(product_id)
GROUP BY PT.vehicle_maker
ORDER BY VehicleCount DESC
LIMIT 5
```

Output:

Showing 5 rows

vehicle_maker	VehicleCount
Chevrolet	125
Ford	97
Toyota	78
Pontiac	75
Dodge	72

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Observations and Insights:

- Chevrolet is the most popular vehicle maker
- 4 of the top 5 vehicle makers are American companies

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT state, vehicle_maker, TotalQuantity, StateRank
FROM (
    SELECT CT.state,
       PT.vehicle_maker,
       SUM(OT.quantity) TotalQuantity,
       RANK() OVER(PARTITION BY CT.state ORDER BY SUM(OT.quantity) DESC) StateRank
    FROM order_t OT
        JOIN product_t PT USING(product_id)
        JOIN customer_t CT USING(customer_id)
    GROUP BY CT.state, PT.vehicle_maker) AS RankedTable
WHERE StateRank = 1;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT state, vehicle_maker, TotalQuantity, StateRank
FROM (
   SELECT CT.state,
     PT.vehicle_maker,
     SUM(OT.quantity) TotalQuantity,
     RANK() OVER(PARTITION BY CT.state ORDER BY SUM(OT.quantity) DESC) StateRank
   FROM order_t OT
      JOIN product_t PT USING(product_id)
      JOIN customer_t CT USING(customer_id)
   GROUP BY CT.state, PT.vehicle_maker) AS RankedTable
WHERE StateRank = 1
```

Output:

Showing first 10 rows out of 88 rows

state	vehicle_maker	TotalQuantity	StateRank
Alabama	Dodge	7	1
Alaska	Chevrolet	3	1
Arizona	Pontiac	6	1
Arkansas	Volkswagen	2	1
Arkansas	Pontiac	2	1

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Observations and Insights:

- Alabama prefers Dodge
- Some states have a tie for the most popular vehicle maker

Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
SELECT AVG(Feedback_Rating) OverallRating
FROM (
    SELECT
        CASE WHEN customer_feedback ='Very Bad' THEN 1
             WHEN customer_feedback ='Bad' THEN 2
             WHEN customer_feedback ='Okay' THEN 3
             WHEN customer_feedback ='Good' THEN 4
             WHEN customer_feedback ='Very Good' THEN 5
        END AS Feedback_Rating
    FROM order_t);
```

```
SELECT quarter_number, AVG(Feedback_Rating) Rating
FROM (
    SELECT quarter_number,
        CASE WHEN customer_feedback ='Very Bad' THEN 1
             WHEN customer_feedback ='Bad' THEN 2
             WHEN customer_feedback ='Okay' THEN 3
             WHEN customer_feedback ='Good' THEN 4
             WHEN customer_feedback ='Very Good' THEN 5
        END AS Feedback_Rating
    FROM order_t)
GROUP BY quarter_number;
```

Output:

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Result: Passed

Query 1

Query:

```
SELECT AVG(Feedback_Rating) OverallRating
FROM (
   SELECT
      CASE WHEN customer_feedback ='Very Bad' THEN 1
          WHEN customer_feedback ='Bad' THEN 2
          WHEN customer_feedback ='Okay' THEN 3
          WHEN customer_feedback ='Good' THEN 4
          WHEN customer_feedback ='Very Good' THEN 5
      END AS Feedback_Rating
   FROM order_t)
```

Output:

Showing 1 rows

OverallRating
3.135

Result: Passed

Query 1

Query:

```
SELECT quarter_number, AVG(Feedback_Rating) Rating
FROM (
   SELECT quarter_number,
      CASE WHEN customer_feedback ='Very Bad' THEN 1
          WHEN customer_feedback ='Bad' THEN 2
          WHEN customer_feedback ='Okay' THEN 3
          WHEN customer_feedback ='Good' THEN 4
          WHEN customer_feedback ='Very Good' THEN 5
      END AS Feedback_Rating
   FROM order_t)
GROUP BY quarter_number
```

Output:

Showing 4 rows

quarter_number	Rating
1	3.554838709677419
2	3.354961832061069
3	2.9563318777292578
4	2.3969849246231156

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Observations and Insights:

- There is a downward trend for the average rating when moving from quarter to quarter
- Customers are happiest on average during the 1st quarter
- Quarters 1 and 2 are above average where as quarters 3 and 4 are below average

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT quarter_number,
       ROUND((SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) * 1.0) /
COUNT(customer_feedback) *100, 2) VeryBadPercentage,
       ROUND((SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) * 1.0) /
COUNT(customer_feedback) *100, 2) BadPercentage,
       ROUND((SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) * 1.0) /
COUNT(customer_feedback) *100, 2) OkayPercentage,
       ROUND((SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) * 1.0) /
COUNT(customer_feedback) *100, 2) GoodPercentage,
       ROUND((SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) * 1.0) /
COUNT(customer_feedback) *100, 2) VeryGoodPercentage
FROM order_t
GROUP BY quarter_number;
```

Output:

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Result: Passed

Query 1

Query:

```
SELECT quarter_number,
    ROUND((SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) * 1.0) / COUNT(customer_feedback)
*100, 2) VeryBadPercentage,
    ROUND((SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) * 1.0) / COUNT(customer_feedback) *100, 2)
BadPercentage,
    ROUND((SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) * 1.0) / COUNT(customer_feedback) *100, 2)
OkayPercentage,
    ROUND((SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) * 1.0) / COUNT(customer_feedback) *100,
2) GoodPercentage,
    ROUND((SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) * 1.0) / COUNT(customer_feedback)
*100, 2) VeryGoodPercentage
FROM order_t
GROUP BY quarter_number
```

Output:

Showing 4 rows

quarter_number	VeryBadPercentage	BadPercentage	OkayPercentage	GoodPercentage	VeryGoodPerc
1	10.97	11.29	19.03	28.71	30
2	14.89	14.12	20.23	22.14	28.63
3	17.9	22.71	21.83	20.96	16.59
4	30.65	29.15	20.1	10.05	10.05

Observations and Insights:

- Very bad and bad percentage was low in quarters 1 and 2 but increased in quarters 3 and 4
- Okay percentage was relatively consistent in each quarter
- Good and very good percentage was high in quarters 1 and 2 but dropped in quarters 3 and 4
- The customers seem to be getting more dissatisfied through each quarter

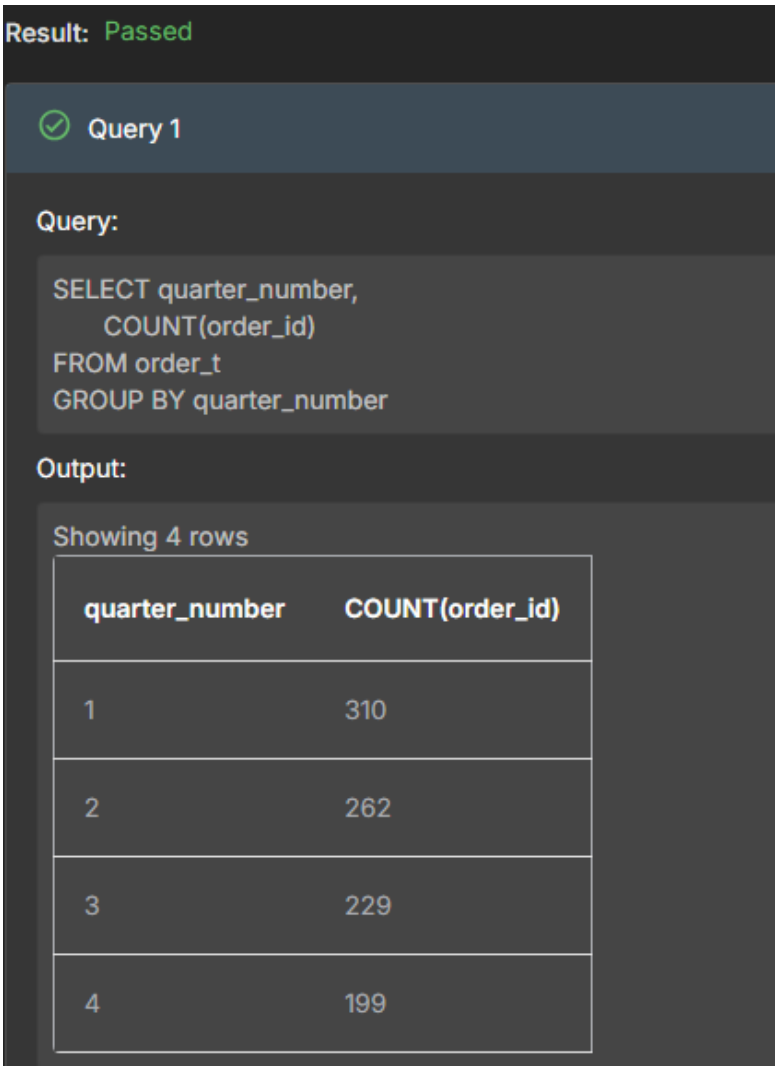
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Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT quarter_number,
       COUNT(order_id)
FROM order_t
GROUP BY quarter_number;
```

Output:



quarter_number	COUNT(order_id)
1	310
2	262
3	229
4	199

Observations and Insights:

- The number of orders drop going from 1 to 2, 2 to 3, and 3 to 4
- Quarter 4 has about a third less orders than quarter 1

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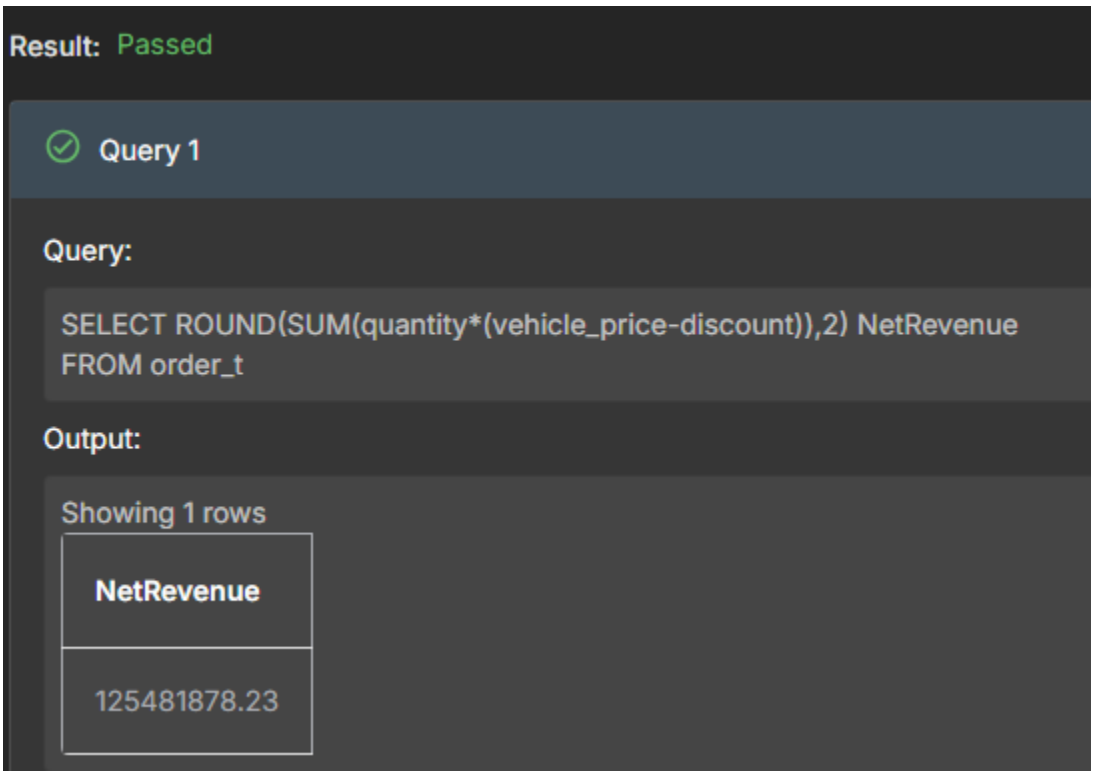
Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
SELECT ROUND(SUM(quantity*(vehicle_price-discount)),2) NetRevenue
FROM order_t;
```

```
SELECT Quarter,
       LAG(NetRevenue,1) OVER() PreviousNetRevenue,
       NetRevenue,
       ROUND((((LAG(NetRevenue) OVER()-NetRevenue)/LAG(NetRevenue,1) OVER())*100),2)
QtrOvrQtrPrctChg
FROM (SELECT quarter_number Quarter,
       ROUND(SUM(quantity*(vehicle_price-discount)),2) NetRevenue
      FROM order_t
      GROUP BY Quarter);
```

Output:



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Result: Passed

Query 1

Query:

```
SELECT Quarter,
      LAG(NetRevenue,1) OVER() PreviousNetRevenue,
      NetRevenue,
      ROUND((((LAG(NetRevenue) OVER()-NetRevenue)/LAG(NetRevenue,1) OVER())*100),2) QtrOvrQtrPrctChg
FROM (SELECT quarter_number Quarter,
      ROUND(SUM(quantity*(vehicle_price-discount)),2) NetRevenue
     FROM order_t
     GROUP BY Quarter)
```

Output:

Showing 4 rows

Quarter	PreviousNetRevenue	NetRevenue	QtrOvrQtrPrctChg
1		39637378.16	
2	39637378.16	32913497.44	16.96
3	32913497.44	29435188.49	10.57
4	29435188.49	23495814.14	20.18

Observations and Insights:

- Couldn't calculate percent change on quarter 1 due to no info about the previous quarter
- Quarter 3 had the smallest quarter over quarter percent change
- Quarter 4 had the largest quarter over quarter percent change

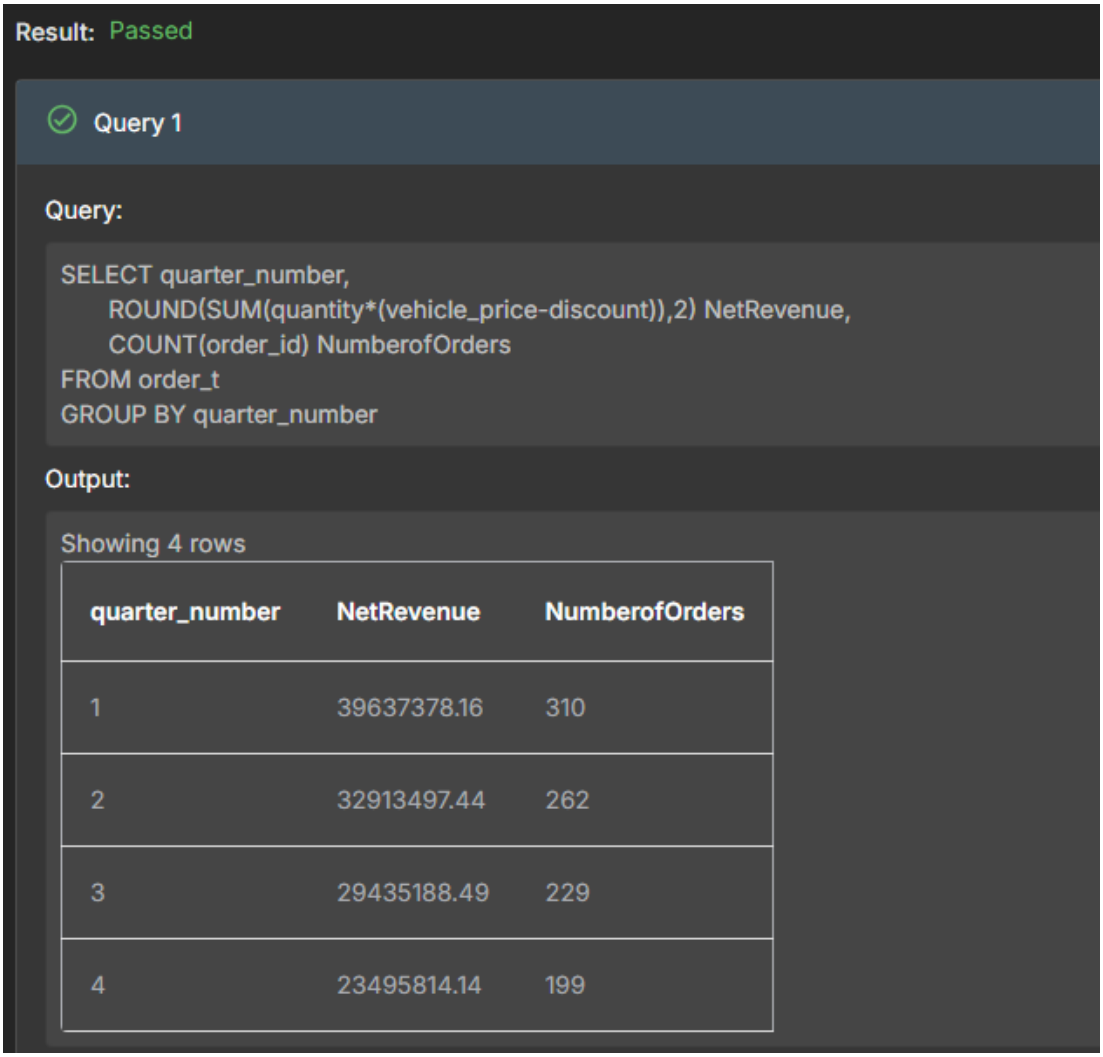
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Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT quarter_number,
       ROUND(SUM(quantity*(vehicle_price-discount)),2) NetRevenue,
       COUNT(order_id) NumberofOrders
FROM order_t
GROUP BY quarter_number;
```

Output:



Observations and Insights:

- The higher the number of orders, the higher the net revenue
- Because the number of orders decreased from quarter to quarter, the net revenue also decreased

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Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT credit_card_type,
       AVG(discount)
FROM customer_t
    JOIN order_t USING(customer_id)
GROUP BY credit_card_type;
```

Output:

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Result: Passed

Query 1

Query:

```
SELECT credit_card_type,
       AVG(discount)
FROM customer_t
   JOIN order_t USING(customer_id)
GROUP BY credit_card_type
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	AVG(discount)
americanexpress	0.616326530612245
bankcard	0.6095454545454548
china-unionpay	0.6221739130434784
diners-club-carte-blan...	0.6144897959183674
diners-club-enroute	0.5997916666666666
diners-club-international	0.584
diners-club-us-ca	0.6146153846153846

Observations and Insights:

- The average discount between different cards seems to be fairly close
- Several cards average discount is about 0.61%
- Diners-club-international had the lowest average discount of cards displayed

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Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT quarter_number,
       julianday(ship_date)-julianday(order_date) Time_Taken_In_Days
FROM order_t
GROUP BY quarter_number;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT quarter_number,
     julianday(ship_date)-julianday(order_date) Time_Taken_In_Days
FROM order_t
GROUP BY quarter_number
```

Output:

Showing 4 rows

quarter_number	Time_Taken_In_Days
1	65
2	123
3	46
4	35

Observations and Insights:

- Quarter 2 had a much larger time from order placed to being shipped than other quarters

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Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
125481878.23	1000	994	3.135
Last Quarter Revenue	**Last quarter Orders**	**Average Days to Ship**	**% Good Feedback**
23495814.14	199	97.964	21.5

Note: These values must be derived using SQL queries. Some of them may have already been obtained while answering previous questions.

Business Recommendations

- Find out why customers are placing fewer orders as each quarter moves on
- Find out why customers are not as happy in quarters 3 and 4
- Urge employees to continue momentum from quarter 1 throughout the year

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